SCHOLARPULSE AI · AUTHENTIC RESEARCH MANUSCRIPT

LoRA: Low-Rank Adaptation of Large Language Models

Authors: Edward J. Hu · Yelong Shen · Phillip Wallis · Zeyuan Allen-Zhu · Yuanzhi Li · Shean Wang · Lu Wang · Weizhu Chen | Venue: ICLR 2022 (cs.CL) | Published: 2021

Abstract

An important paradigm of natural language processing consists of large-scale pre-training on general domain data and adaptation to particular tasks. As we pre-train larger models, full fine-tuning, which retrains all model parameters, becomes less feasible. We propose Low-Rank Adaptation, or LoRA, which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture, greatly reducing the number of trainable parameters for downstream tasks.

4 Method

A neural network contains many dense layers which perform matrix multiplication. The weight matrices in these layers typically have full-rank. When adapting to a specific task, Aghajanyan et al. (2020) shows that the pre-trained language models have a low "intrinsic dimension". Inspired by this, we hypothesize the updates to the weights also have a low "intrinsic rank" during adaptation. For a pre-trained weight matrix W0 in R^(d x k), we constrain its update by representing the latter with a low-rank decomposition W0 + Delta W = W0 + BA, where B in R^(d x r), A in R^(r x k), and the rank r << min(d, k).